

NEPA Project Analysis: Clean Energy Environmental Reviews

National Environmental Policy Act Text Corpus (NEPATEC) 2.0 Analysis

Your Name

2026-02-25

Table of contents

1 Project Overview	1
1.1 Project context: National Environmental Policy Act (NEPA)	1
1.2 Data structure	2
2 Project deliverables	5
2.1 Phase 1	5
2.2 Phase 2	6
3 Project deliverable timelines	6
4 Project Structure	7

1 Project Overview

This document is a good reference for how to contextualize this project, the project goals and data, and to clearly understand the project deliverables and timeline.

1.1 Project context: National Environmental Policy Act (NEPA)

Federal environmental permitting—and, relatedly, the National Environmental Policy Act (NEPA)—has often been blamed as a core contributor to delays in infrastructure deployment. However, research to date on NEPA has been hindered by federal agency data management

practices: NEPA documents are scattered across numerous agency databases, often in machine-unreadable formats, and typically lack basic metadata and other identifiers. As such, researchers have had to craft bespoke subsets of NEPA data from which to glean insights—a time-consuming task that has limited the information available about NEPA’s effectiveness and paved the way for a national permitting reform conversation rife with anecdotes and cherry-picked information.

The newly released National Environmental Policy Act Text Corpus (NEPATEC) 2.0 dataset from the Pacific Northwest National Laboratory’s (PNNL’s) PermitAI project has the potential to add new, more comprehensive evidence into this conversation. PNNL has built and released a comprehensive dataset of past environmental reviews and permitting documents containing millions of pages and billions of words in machine-readable JSON format. The database contains more than 120,000 NEPA documents¹ from 60,000 projects prepared by more than 60 different agencies. Each document contains metadata for (as applicable):

- Lead agency
- Category
- Type of review
- Name of project
- Location
- Project sponsor
- Project sector
- Project type (a subset of project sector)
- Type of document
- Document title
- Agency or contractor responsible for preparing the document
- Categorical exclusion category
- Summary of the proposed action

Note

The National Environmental Policy Act Text Corpus (NEPATEC) 2.0 can be accessed [here](#) on huggingface.

1.2 Data structure

This shows the original data structure of the project in its original json format. The code in this repo manipulates this raw data.

¹categorical exclusions, draft and final environmental assessments, draft and final environmental impact statements, records of decision, findings of no significant impact, and other supporting documentation

```

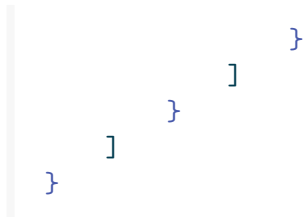
{
  "project": {
    "project_ID": "UNIQUE PROJECT ID FOR PUBLIC VERSION",
    "project_title": {
      "value": ""
    },
    "project_sector": {
      "value": ""
    },
    "project_type": {
      "value": ""
    },
    "project_description": {
      "value": ""
    },
    "project_sponsor": {
      "value": ""
    },
    "location": {
      "value": ""
    }
  },
  "process": {
    "process_family": {
      "value": ""
    },
    "process_type": {
      "value": ""
    },
    "lead_agency": {
      "value": ""
    }
  },
  "documents": [
    {
      "metadata": {
        "document_metadata": {
          "document_ID": {
            "value": "UNIQUE DOC/FILE ID FOR PUBLIC VERSION"
          },
          "document_type": {

```

```

        "value": ""
    },
    "document_title": {
        "value": ""
    },
    "prepared_by": {
        "value": ""
    },
    "ce_category": {
        "value": ""
    }
},
"file_metadata":{
    "file_ID": {
        "value": "UNIQUE DOC/FILE ID FOR PUBLIC VERSION"
    },
    "file_name": {
        "value": "PDF NAME"
    },
    "section_or_volume_title": {
        "value": ""
    },
    "main_document": {
        "value": ""
    },
    "total_pages": {
        "value": ""
    },
    "file_provider": {
        "value": ""
    }
}
},
"pages": [
    {
        "page number": 1,
        "page text": "PAGE 1 TEXT"
    },
    {
        "page number": 2,
        "page text": "PAGE 2 TEXT"
    }
]

```



2 Project deliverables

For this project, we want to create tables, figures, and maps that help us learn about the data and answer the following questions:

2.1 Phase 1

Table 1: Phase 1 Deliverable Timeline

Deliverable	Due_Date
1. Clean Energy Projects	Jan 23, 2026
2. Programmatic Reviews	Feb 6, 2026
3. CE vs EA vs EIS	Jan 23, 2026
4. Geography	Feb 6, 2026
5. Pages Over Time	Feb 27, 2026
6. Technology-Specific	Feb 27, 2026

1. **Data on number of clean energy projects within the dataset:** number of projects broken down by technology (e.g., offshore and onshore wind, solar, geothermal, nuclear), lead agency, and location
2. **Data on programmatic and tiered reviews:** how many tiered reviews are there compared to total and are they completed faster
3. **Data on how many clean energy projects have been categorically excluded vs. have required environmental assessments and environmental impact statements**
 - Broken out by number of projects, generation capacity, and change over time
4. **Data on geography/project location:** whether projects are multi-state or multi-agency
5. **Number of pages over time,** including pre- and post- Fiscal Responsibility Act of 2023 (FRA), which set page limit requirements

6. Technology-specific inquiries

- Transmission lines: length of lines from project summary correlated with timelines, location, etc.
- Geothermal: timelines of environmental reviews for different phases of the same project
- Carbon and hydrogen pipelines (if available): length of pipelines from project summary correlated with timelines, location, etc.; compare to natural gas pipelines

2.2 Phase 2

Phase 2 of the project is still not settled and we'll need to scope it based on what we find and do in Phase 1. Nevertheless, these are the envisioned deliverables:

1. **Reasons why NEPA was triggered** (e.g., federal land, federal funding) for different types of projects
2. **Determinations of significance across resource areas**; factors that contribute to a determination of “significant impact”
 - Starting with mitigated FONSI
3. **Differences and similarities between NEPA reviews** for fossil fuel and clean energy projects, as well as linear and non-linear projects—application of categorical exclusions, timelines, geography, etc.
4. **Timelines for categorical exclusions, environmental assessments, and environmental impact statements**, including segmentation by years (e.g., pre- and post-FRA [which set timelines for reviews], different CEQ NEPA regulations, agency, and type of project)
 - Timeline outliers could then be investigated through a case study approach to identify contributing factors, including whether NEPA was a cause of delay or not
 - May need to cross-reference with the Notice of Intent in the Federal Register using their API to get the start date
5. **Technical support for new regulatory categorical exclusion development**: identifying patterns in FONSI

3 Project deliverable timelines

Table 2: Project Timeline and Deliverables

Meeting	Date	Deliverables
Kickoff	Jan 9, 2026	(0) Build database
1	Jan 23, 2026	(1) Clean energy projects(3) CE vs EA vs EIS
2	Feb 6, 2026	(2) Reviews(4) Geography
3	Feb 27, 2026	(5) Pages(6) Technology
4	Mar 6, 2026	Present all findings

4 Project Structure

This project is organized into the following directories:

- **code/**: Python scripts for data processing and analysis pipelines
- **data/**: Raw and processed data files (parquet format) organized by review type (EA, EIS, CE)
- **notebooks/**: Jupyter notebooks for exploratory analysis
- **reports/**: Generated reports and deliverable documents
- **output/**: Analysis outputs including tables, figures, and visualizations
- **notes/**: Internal project documentation and working notes
- **literature/**: Reference materials and background documents